

# AI-POWERED WORKFORCE ANALYTICS & TALENT INTELLIGENCE DASHBOARD

## MILESTONE 3 · Individual Task Documentation

| Item                           | Detail                                                                                           |
|--------------------------------|--------------------------------------------------------------------------------------------------|
| Team Member                    | Swetha Sneharaj                                                                                  |
| Assigned Task / Responsibility | Engagement Analytics Module — Develop visualization for employee engagement trends.              |
| Key Activities                 | Display engagement scores and trends; create department-wise and time-based engagement analysis. |

### 1. Objective of My Module

The Engagement Analytics Module turns the five satisfaction fields in the workforce dataset into a single, business-readable engagement measure, exposes it on the Workforce Intelligence Dashboard, and connects it to the outcome of primary interest to HR — whether an employee stays.

The scope of this module is the engagement measure itself, the visuals that present it, and the data contract that lets those visuals sit inside the shared dashboard file without conflicting with the other analytics modules.

### 2. Method — The Engagement Index

The dataset contains no engagement score. It contains five separate 1–4 satisfaction scales. The module combines them into one index so that engagement can be filtered, banded and compared like any other KPI.

**Definition:** the mean of five 1–4 scales, rescaled to 0–100 as  $(\text{mean} - 1) \div 3 \times 100$ .

- Job Satisfaction
- Environment Satisfaction
- Job Involvement
- Relationship Satisfaction
- Work-Life Balance

**PerformanceRating is deliberately excluded.** It takes only the values 3 and 4, with 1,244 of 1,470 employees (84.6%) at 3. Including it would add almost no variance and would compress every employee toward the same score.

Bands: Low below 45 · Moderate 45–69 · High 70 and above.

### 3. Findings

#### 3.1 Engagement distribution

Mean 57.69, median 60.0, standard deviation 14.28. The workforce is concentrated in the Moderate band: 957 employees (65.1%), against 226 Low (15.4%) and 287 High (19.5%).

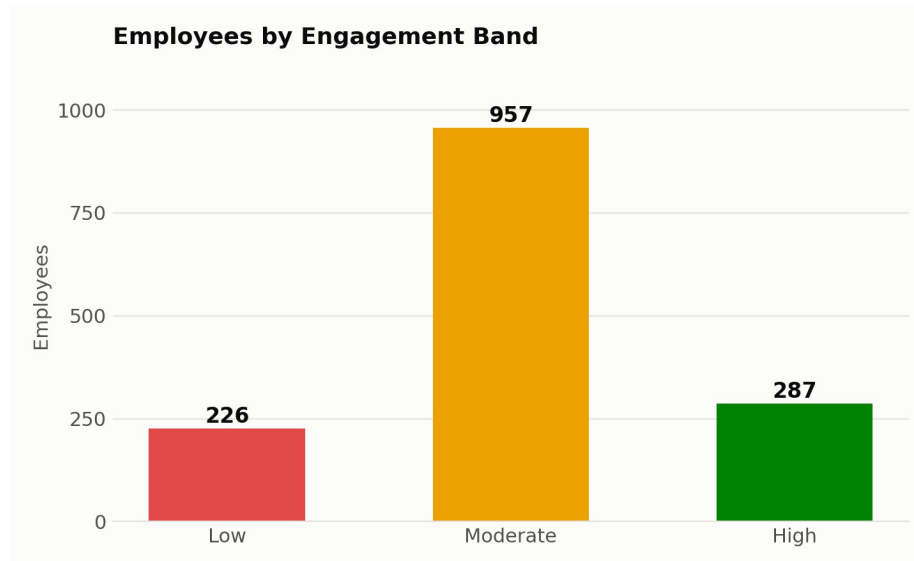


Figure 1 — Employees by engagement band.

#### 3.2 Engagement is strongly associated with attrition

Engagement separates leavers from stayers strongly and significantly. Employees who left average 51.39 on the index; employees who stayed average 58.90 — a gap of 7.51 index points, Cohen's  $d = 0.536$ , Welch t-test  $p = 1.5e-11$ .

| Engagement band    | Headcount | Leavers | Attrition rate |
|--------------------|-----------|---------|----------------|
| Low (index < 45)   | 226       | 72      | 31.9%          |
| Moderate (45–69)   | 957       | 137     | 14.3%          |
| High ( $\geq 70$ ) | 287       | 28      | 9.8%           |
| All employees      | 1,470     | 237     | 16.1%          |

Chi-square across the three bands:  $\chi^2 = 52.29$ ,  $p = 4.4e-12$ .

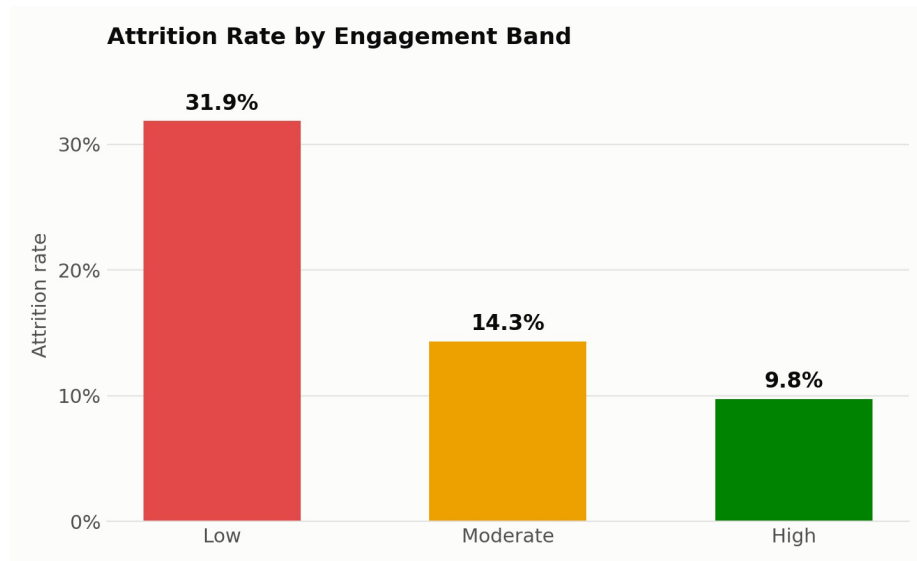


Figure 2 — Attrition rate by engagement band.

### 3.3 Driver ranking

Ranking the five drivers by how far each separates leavers from stayers shows that the index is not carried evenly by its components.

| Driver                    | Stayers | Leavers | Gap   | p-value |
|---------------------------|---------|---------|-------|---------|
| Job Satisfaction          | 59.29   | 48.95   | 10.34 | 0.0001  |
| Environment Satisfaction  | 59.04   | 48.80   | 10.24 | 0.0002  |
| Job Involvement           | 59.02   | 50.63   | 8.38  | 4.7e-06 |
| Relationship Satisfaction | 57.80   | 53.31   | 4.49  | 0.090   |
| Work-Life Balance         | 59.37   | 55.27   | 4.09  | 0.031   |

The first three drivers account for most of the separation. Relationship Satisfaction does not reach statistical significance ( $p = 0.090$ ), and Work-Life Balance separates the two groups by approximately half as much as the leading drivers. Retention measures should therefore target job content and work environment rather than relationship or work-life-balance interventions.

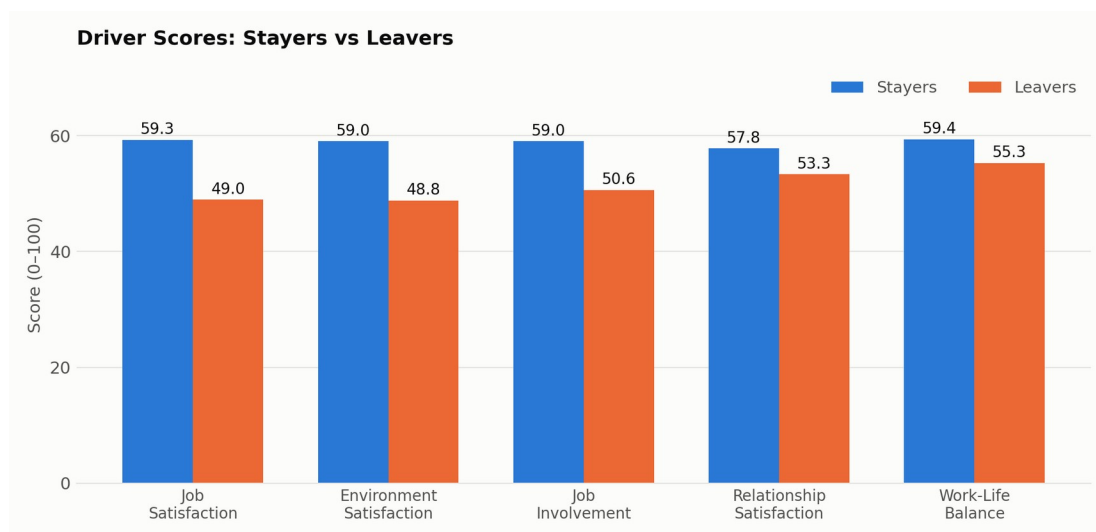


Figure 3 — Driver scores, stayers versus leavers.

## 4. Department-wise and Time-based Analysis

Department-wise and time-based engagement analysis were both performed. Both returned negative results. These are reported here as findings rather than presented as trends.

### 4.1 There is no significant engagement variation across organisational segments

| Segment cut       | ANOVA p | Variance explained | Spread   | Verdict              |
|-------------------|---------|--------------------|----------|----------------------|
| Department        | 0.78    | 0.034%             | 1.32 pts | Not significant      |
| Tenure band       | 0.87    | 0.087%             | 1.04 pts | Not significant      |
| Job role          | 0.95    | 0.193%             | 2.16 pts | Not significant      |
| Business travel   | 0.51    | 0.091%             | 1.06 pts | Not significant      |
| Age band          | 0.40    | 0.200%             | 2.14 pts | Not significant      |
| Gender            | 0.19    | 0.115%             | 0.99 pts | Not significant      |
| Attrition outcome | 7.5e-14 | 3.74%              | 7.51 pts | Strongly significant |

No organisational cut shows significant variation. Between-department spread is 1.32 index points against a within-group standard deviation of 14.28 — an order of magnitude smaller than the variation within each department. The only cut that separates is the attrition outcome.

This reflects the structure of the source data: the satisfaction fields carry almost no variation across organisational segments.

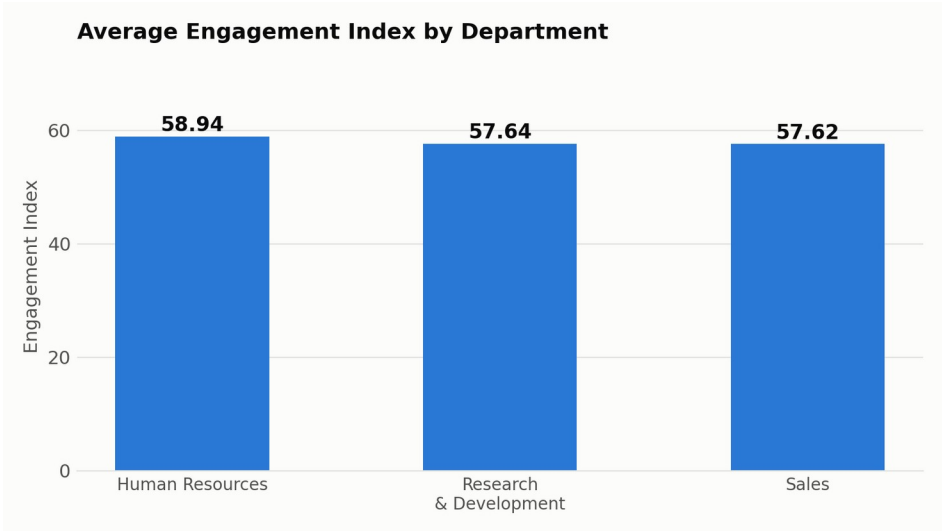


Figure 4 — Average engagement index by department. The y-axis is anchored at zero.

### 4.2 The dataset contains no date column

No true time-based analysis is possible. The dataset contains no date, month, quarter or survey-period column. Any trend analysis is therefore a cross-sectional tenure proxy: a comparison of employees who currently have different lengths of service, measured at a single point in time.

Tenure is therefore presented as a labelled proxy, paired with attrition across the same segmentation for comparison. Engagement does not vary across tenure ( $p = 0.87$ ). Attrition does: it runs at 29.8% in the first two years and falls to 6.7% at 11–20 years.

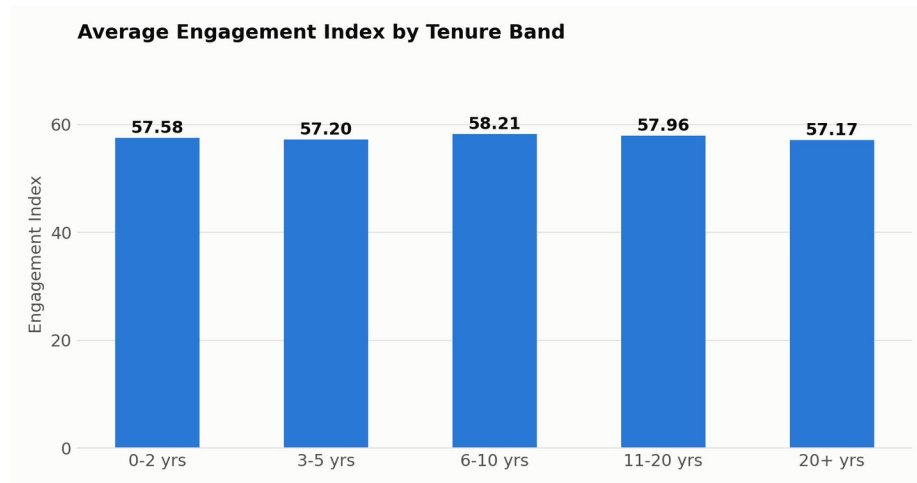


Figure 5 — Average engagement index by tenure band.

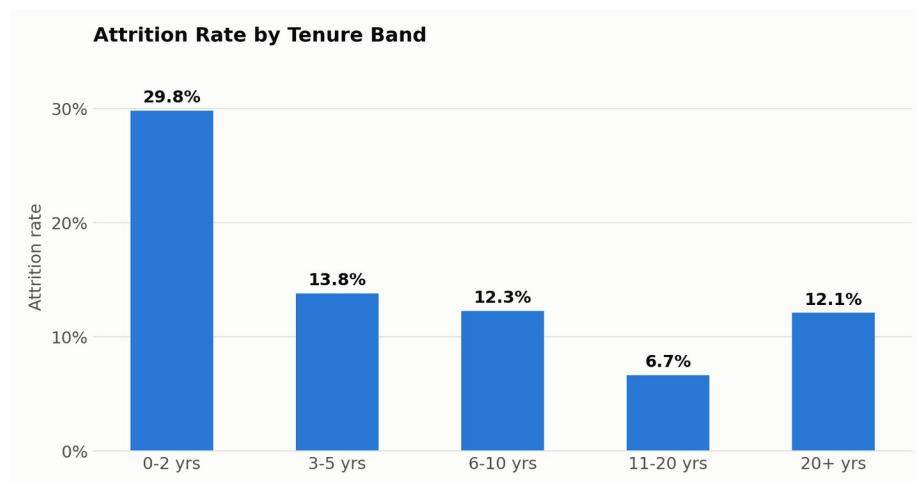


Figure 6 — Attrition rate by tenure band, across the same segmentation as Figure 5.

## 5. Overtime, Engagement and Attrition

Employees working overtime report engagement 1.99 points higher than those who do not (59.12 against 57.13,  $p = 0.014$ ), yet they leave at 30.5% against 10.4% — nearly three times the rate.

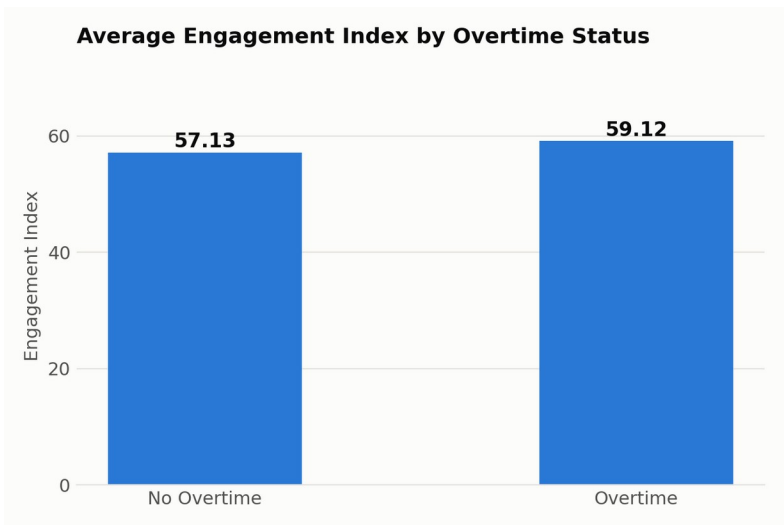


Figure 7 — Average engagement index by overtime status.

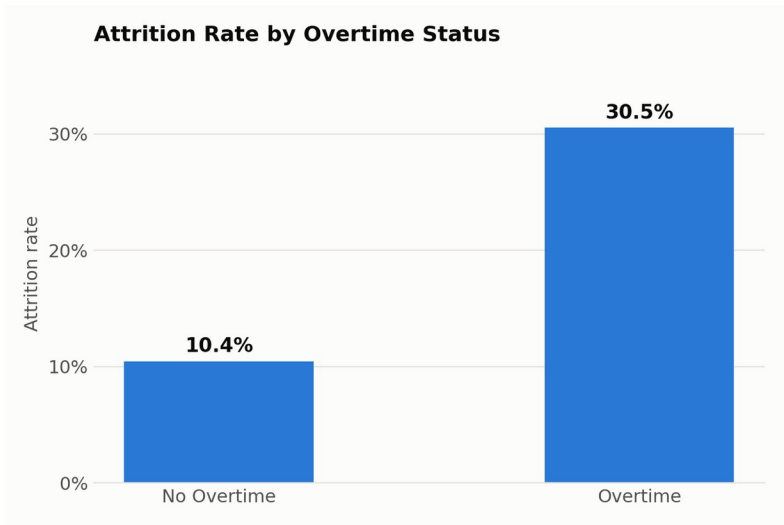


Figure 8 — Attrition rate by overtime status.

This has a direct implication for interpretation: a satisfaction survey does not capture exit intent. An employee can report being engaged and still leave. Engagement should therefore be presented as one risk input among several rather than as a standalone retention measure.

This connects directly to Milestone 2, where OverTime was among the strongest predictors in the attrition model.

## 6. Summary of Findings

| Question                                | Result                                                                              |
|-----------------------------------------|-------------------------------------------------------------------------------------|
| What is the workforce engagement level? | Mean index 57.69 (SD 14.28). 65.1% sit in the Moderate band, 15.4% Low, 19.5% High. |

| Question                                                     | Result                                                                                                                                               |
|--------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------|
| Does engagement vary by department?                          | No. ANOVA $p = 0.78$ ; department explains 0.034% of the variance, a 1.32-point spread against a within-group SD of 14.28.                           |
| Does engagement vary by tenure, role, travel, age or gender? | No. Every cut returns $p > 0.19$ and explains under 0.21% of the variance.                                                                           |
| Can engagement be tracked over time?                         | No. The dataset contains no date column. Tenure band is a cross-sectional proxy only.                                                                |
| Is engagement related to attrition?                          | Yes, strongly. Leavers average 51.39 against 58.90 for stayers — a 7.51-point gap, Cohen's $d = 0.536$ , $p = 1.5e-11$ .                             |
| How large is the difference in practice?                     | Low-band employees leave at 31.9% against 9.8% in the High band — a 3.3x risk multiple. $\chi^2 = 52.29$ , $p = 4.4e-12$ .                           |
| Which drivers carry the signal?                              | Job Satisfaction (10.34), Environment Satisfaction (10.24) and Job Involvement (8.38). Relationship Satisfaction is not significant ( $p = 0.090$ ). |
| Does overtime lower engagement?                              | No — overtime employees score 1.99 points higher ( $p = 0.014$ ), yet leave at 30.5% against 10.4%.                                                  |